1 Set Theory & Maps

Definition 1.1 (Set). A *set* is a collection of distinct objects (its *elements*). $x \in M$: x is an element of M; $x \notin M$: not. The *empty set* is $\emptyset$.

Definition 1.2 (Set operations). $A \subseteq B$ iff every element of A lies in B; $A = B$ iff $A \subseteq B$ and $B \subseteq A$.

$$A \cap B := \{x \mid x \in A \text{ and } x \in B\}, \quad A \cup B := \{x \mid x \in A \text{ or } x \in B\},$$
$$A \setminus B := \{x \mid x \in A,\, x \notin B\}.$$

$A \times B := \{(a,b) \mid a \in A,\, b \in B\}$ (Cartesian product); $A^n = A \times \cdots \times A$ (n factors).

Definition 1.3 (Map). A *map* (function) $f : X \to Y$ assigns to each $x \in X$ exactly one value $f(x) \in Y$ (X: domain, Y: codomain). *Image* $f(A) = \{f(x) \mid x \in A\}$; *preimage* $f^{-1}(B) = \{x \in X \mid f(x) \in B\}$.

Definition 1.4 (Injective / surjective / bijective). f is *injective* if $f(x) = f(x') \Rightarrow x = x'$; *surjective* if $f(X) = Y$; *bijective* if both.

Theorem 1.1 (Inverse map). *$f : X \to Y$ has an inverse $f^{-1} : Y \to X$ (with $f^{-1} \circ f = \mathrm{id}_X$ and $f \circ f^{-1} = \mathrm{id}_Y$) iff f is bijective; then f^{-1} is unique.*

Definition 1.5 (Composition). $(g \circ f)(x) = g(f(x))$ for $f : X \to Y$, $g : Y \to Z$.

Example 1.1. Dirichlet function $f(x) = 1$ if $x \in \mathbb{Q}$, else 0: a valid map, continuous nowhere.

2 Number Systems & Inequalities

Definition 2.1 (Group / ring / field). *Group* $(G, \cdot)$: associative, identity 1, inverses; *abelian* if commutative. *Ring* $(R, +, \cdot)$: $(R, +)$ abelian group, $\cdot$ associative with 1, distributive. *Field* K: commutative ring with $1 \neq 0$ and every $x \neq 0$ invertible.

Example 2.1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ abelian groups; $(\mathbb{Q} \setminus \{0\}, \cdot)$, $(\mathbb{R} \setminus \{0\}, \cdot)$ abelian groups. $\mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ are subrings; $\mathbb{Q}, \mathbb{R}$ are fields.

Theorem 2.1 (Induction principle). *If A_0 holds and $A_n \Rightarrow A_{n+1}$ for all $n \in \mathbb{N}$, then all A_n hold. Variant: if $A_0, \ldots, A_{n-1} \Rightarrow A_n$ for all n, then all hold.*

Definition 2.2 (Order axioms of $\mathbb{R}$). For each $x \in \mathbb{R}$ exactly one of $x > 0$, $x = 0$, $-x > 0$ holds; $x, y > 0 \Rightarrow x + y > 0$ and $xy > 0$; *Archimedes*: for every $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n - x > 0$.

Proposition 2.1 (Inequality rules). *$x < y$, $y < z \Rightarrow x < z$; $a < b$, $x < y \Rightarrow a + x < b + y$; $a > 0$, $x < y \Rightarrow ax < ay$ (reverses for $a < 0$); $x \neq 0 \Rightarrow x^2 > 0$; $0 < x < y \Rightarrow x^{-1} > y^{-1}$; for every $\varepsilon > 0$ there is $n > 0$ with $1/n < \varepsilon$.*

Theorem 2.2 (Bernoulli's inequality). *$(1+x)^n \geq 1 + nx$ for $x \geq -1$, $n \in \mathbb{N}$.*

Sketch. *Induction on n: multiply $(1+x)^n \geq 1 + nx$ by $(1+x) \geq 0$.*

Theorem 2.3 (Density of $\mathbb{Q}$). *Between any two distinct reals $x < y$ there is a rational r: $x < r < y$.*

Sketch. *Pick q with $1/q < y - x$; set $p = \lfloor qx \rfloor + 1$; then $x < p/q < y$.*

Definition 2.3 (Absolute value). $|x| = x$ for $x \geq 0$, $-x$ for $x < 0$.

Theorem 2.4 (Triangle inequality). *$|x \pm y| \leq |x| + |y|$; also $\big||x| - |y|\big| \leq |x + y|$.*

Definition 2.4 (Floor / ceiling). $\lfloor x \rfloor$ = largest integer $\leq x$; $\lceil x \rceil$ = smallest integer $\geq x$.

Definition 2.5 (Supremum / infimum). $s = \sup A$ iff s is an upper bound of A and no $t < s$ is one; $i = \inf A$ dually. *Completeness of* $\mathbb{R}$: every nonempty set bounded above has a supremum.

3 Sequences

Definition 3.1 (Sequence). A *sequence* in Y is a map $x : \mathbb{N} \to Y$, written $(x_n)_{n \in \mathbb{N}}$ or $(x_n)_{n=0}^{\infty}$.

Definition 3.2 (Convergence). $(x_n) \to a$ (*converges to* a) iff for every $\varepsilon > 0$ there is $D \in \mathbb{N}$ with $|x_n - a| < \varepsilon$ for all $n > D$. Notation: $\lim_{n \to \infty} x_n = a$. Otherwise *divergent*.

Example 3.1. $1/n \to 0$; $(-1)^n$ diverges; $x^n \to 0$ for $|x| < 1$, $x^n \to 1$ for $x = 1$, else diverges (geometric sequence).

Definition 3.3 (Bounded). (x_n) is *bounded* iff $\{x_n \mid n \in \mathbb{N}\}$ is bounded, i.e. there is $c \in \mathbb{R}$ with $|x_n| \leq c$ for all n.

Lemma 3.1. *Every convergent sequence is bounded.*

Lemma 3.2 (Uniqueness of limits). *A convergent sequence has exactly one limit.*

Sketch. *If $x_n \to a$ and $x_n \to b$, then $|a - b| \leq |a - x_n| + |x_n - b| < \varepsilon$ for all $\varepsilon > 0$, hence $a = b$.*

Proposition 3.1 (Limit rules). *If $x_n \to a$, $y_n \to b$: $x_n + y_n \to a + b$; $x_n y_n \to ab$; if $b \neq 0$ then $x_n / y_n \to a/b$ (for large n); if $x_n \leq y_n$ for all n then $a \leq b$.*

Theorem 3.1 (Ratios of polynomials). *For polynomials f (deg d) and $g \neq 0$ (deg e): $\lim_{n \to \infty} f(n)/g(n) = a_d / b_e$ if $d = e$; $= 0$ if $d < e$; diverges if $d > e$.*

Sketch. *Factor out n^d, n^e; use $1/n \to 0$.*

Theorem 3.2 (Squeeze theorem). *If $x_n \leq y_n \leq z_n$ for all n and $x_n, z_n \to a$, then $y_n \to a$.*

4 Cauchy Sequences & Completeness

Definition 4.1 (Cauchy sequence). (x_n) is a *Cauchy sequence* iff for every $\varepsilon > 0$ there is D with $|x_{n+k} - x_n| < \varepsilon$ for all $k \in \mathbb{N}$, $n > D$.

Lemma 4.1. *Convergent $\Rightarrow$ Cauchy.*

Sketch. *$|x_m - x_n| \leq |x_m - a| + |a - x_n| < 2\varepsilon$.*

Theorem 4.1 (Completeness axiom). *Every Cauchy sequence of real numbers converges to a real limit. (This distinguishes $\mathbb{R}$ from $\mathbb{Q}$.)*

Definition 4.2 (Monotone / subsequence). (x_n) *increasing* if $x_n \leq x_{n+1}$ for all n (*strictly* if $<$); *decreasing* dually. A *subsequence* is $(x_{n_k})_{k \in \mathbb{N}}$ with (n_k) strictly increasing in $\mathbb{N}$.

Theorem 4.2. *Every real sequence has a monotone subsequence.*

Sketch. *Call m "convenient" if $x_n \leq x_m$ for all $n \geq m$. Infinitely many convenient indices give a decreasing subsequence; otherwise build a strictly increasing one inductively.*

Theorem 4.3. *A monotone sequence converges iff it is bounded.*

Sketch. *Convergent $\Rightarrow$ bounded is Lemma above. Conversely, if increasing and not convergent, it is not Cauchy: extract a subsequence with $x_{n_{j+1}} - x_{n_j} \geq \varepsilon$, hence unbounded.*

Example 4.1 (Heron's iteration for $\sqrt{2}$)**.** $x_0 = 1$, $x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n}$. Then $x_n^2 \geq 2$, the sequence is decreasing for $n \geq 1$, hence convergent; the limit w satisfies $w = w/2 + 1/w$, i.e. $w = \sqrt{2}$. This constructs $\sqrt{2}$ as a limit.

Theorem 4.4 (Bolzano–Weierstrass)**.** *Every bounded sequence has a convergent subsequence.*

Sketch. *Take a monotone subsequence (previous theorems); it is bounded, hence converges.*

Remark 4.1. A subsequence of a convergent sequence converges to the same limit. A sequence converges to a iff every subsequence converges to a.

5 Series

Definition 5.1 (Series)**.** The *series* $\sum_{n=0}^{\infty} x_n$ converges iff the sequence of *partial sums* $s_m = \sum_{n=0}^{m} x_n$ converges; then $\sum_{n=0}^{\infty} x_n := \lim_m s_m$.

Example 5.1 (Geometric series)**.** $\sum_{n=0}^{\infty} x^n$ converges iff $|x| < 1$, with sum $\frac{1}{1-x}$.

Sketch. $s_m = \frac{1-x^{m+1}}{1-x}$ for $x \neq 1$; use $x^{m+1} \to 0$ iff $|x| < 1$.

Theorem 5.1 (Cauchy criterion for series)**.** $\sum x_n$ *converges iff for every* $\varepsilon > 0$ *there is* D *with* $|\sum_{n=m+1}^{m+k} x_n| < \varepsilon$ *for all* $m > D$, $k \in \mathbb{N}$.

Lemma 5.1 (Null sequence test)**.** *If* $\sum x_n$ *converges then* $x_n \to 0$. *(Converse false!)*

Example 5.2 (Harmonic series)**.** $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges although $1/n \to 0$.

Sketch. $\sum_{n=m+1}^{2m} \frac{1}{n} \geq m \cdot \frac{1}{2m} = \frac{1}{2}$ violates the Cauchy criterion.

Proposition 5.1 (Rearrangement rules)**.** *Linear combinations:* $\sum(x_n + y_n) = \sum x_n + \sum y_n$, $\sum \lambda x_n = \lambda \sum x_n$. *Adding/removing finitely many terms, or inserting/removing zeros, does not affect convergence. In a convergent series, arbitrary grouping (parenthesizing) of terms is allowed.*

Example 5.3 (Rearrangement matters)**.** The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converges (Leibniz), but a suitable rearrangement converges to $\frac{3}{2}$ times its sum. Conditionally convergent series can be rearranged to any prescribed sum (or to diverge).

Definition 5.2 (Absolute convergence)**.** $\sum x_n$ is *absolutely convergent* iff $\sum |x_n|$ converges.

Proposition 5.2. *Absolute convergence* $\Rightarrow$ *convergence (Cauchy criterion + triangle inequality). Sums and multiples of absolutely convergent series are absolutely convergent.*

Theorem 5.2 (Comparison test (majorant/minorant))**.** *If* $0 \leq |x_n| \leq \mu_n$ *and* $\sum \mu_n$ *converges, then* $\sum x_n$ *converges absolutely. If* $|x_n| \geq \mu_n \geq 0$ *and* $\sum \mu_n$ *diverges, then* $\sum x_n$ *does not converge absolutely.*

Theorem 5.3 (Root test)**.** *If* $\sqrt[n]{|x_n|} \leq q < 1$ *for all* n *(or* $\lim_n \sqrt[n]{|x_n|} < 1$*), then* $\sum x_n$ *converges absolutely.*

Theorem 5.4 (Ratio test)**.** *If* $x_n \neq 0$ *and* $|x_{n+1}/x_n| \leq q < 1$ *for all* n *(or the limit is* < 1*), then* $\sum x_n$ *converges absolutely.*

Remark 5.1. If the root/ratio limit equals 1, no conclusion. If it is > 1, the terms do not tend to 0, so the series diverges.

Theorem 5.5 (Leibniz criterion)**.** *If* (x_n) *decreases monotonically to* 0*, then* $\sum_{n=0}^{\infty} (-1)^n x_n$ *converges.*

Sketch. *Even and odd partial sums are monotone and bounded, with difference* $x_{2m} \to 0$.

Example 5.4. $\sum \frac{1}{n^2}$ converges (compare with $\sum \frac{1}{n(n-1)} = 1$); more generally, $\sum \frac{1}{n^d}$ converges iff $d > 1$. The series $\sum \frac{x^n}{n!}$, $\sum \frac{(-1)^n x^{2n}}{(2n)!}$, $\sum \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ converge absolutely for every x (ratio test) ŏ2014 these define exp, cos, sin.

Theorem 5.6 (Rearrangement theorem)**.** *Every rearrangement of an absolutely convergent series converges absolutely to the same sum.*

Sketch. *For nonnegative terms, the sum equals* sup *of all finite partial sums, independent of order; split a general series into positive and negative parts.*

Theorem 5.7 (Cauchy product)**.** *If* $\sum_j x_j$ *and* $\sum_k y_k$ *converge absolutely, then* $\sum_n \left(\sum_{j+k=n} x_j y_k\right)$ *converges absolutely to* $\left(\sum_j x_j\right)\left(\sum_k y_k\right)$.

6 Countability

Definition 6.1 (Countable)**.** X is *countable* iff $X = \emptyset$ or there is a surjection $\mathbb{N} \to X$.

Proposition 6.1. *Subsets, images of countable sets are countable; finite products and countable unions of countable sets are countable.*

Example 6.1. $\mathbb{Z}$, $\mathbb{Z}^n$, $\mathbb{Q}$, $\mathbb{Q}^n$, and the set of polynomials with rational coefficients are countable.

Theorem 6.1 (Cantor)**.** $\mathbb{R}$ *is not countable.*

Sketch. *Diagonal argument on decimal expansions of* $[0, 1)$.

7 Continuous Functions

Definition 7.1 (Continuity)**.** $f : X \to \mathbb{R}$ ($X \subseteq \mathbb{R}$) is *continuous at* $a \in X$ iff for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(a)| < \varepsilon$ for all $x \in X$ with $|x - a| < \delta$. f is *continuous* iff continuous at every point.

Example 7.1. Constants and id are continuous. The step function (0 for $x \leq 0$, 1 for $x > 0$) is discontinuous at 0. The Dirichlet function is continuous nowhere. $f(x) = |x|$ is continuous but not differentiable at 0.

Proposition 7.1 (Algebra of continuous functions)**.** *Sums, products, quotients (where the denominator is nonzero), and compositions of continuous functions are continuous.*

Theorem 7.1. *Every polynomial is continuous; every rational function is continuous on its domain* $\{t \mid g(t) \neq 0\}$.

Theorem 7.2 (Sequential criterion)**.** f *is continuous at* a *iff for every sequence* (x_n) *in* X *with* $x_n \to a$ *one has* $f(x_n) \to f(a)$.

Sketch. *ε-δ $\Rightarrow$ sequential is direct. Conversely, if f is discontinuous at a, pick "witnesses" x_n with $|x_n - a| < 1/n$ but $|f(x_n) - f(a)| \geq \varepsilon$.*

Remark 7.1. Continuity is a *local* property: if f coincides with a continuous function on a neighborhood of a, then f is continuous at a.

8 Continuous Functions on Intervals

Definition 8.1 (Intervals)**.** Intervals are exactly the subsets $X \subseteq \mathbb{R}$ with the *intermediate value property*: $\alpha, \beta \in$

X, $\alpha < t < \beta \Rightarrow t \in X$. Types: $[a,b]$ closed/compact, (a,b) open, $[a,b)$, $(a,b]$, $[a,\infty)$, $(-\infty,b]$, $(-\infty,\infty) = \mathbb{R}$. "Compact" = closed and bounded.

Theorem 8.1 (Intermediate value theorem)**.** *If I is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval. In particular, if $f(a) < t < f(b)$, there is s between a and b with $f(s) = t$.*

Sketch. *Set $s = \inf\{x \in [a,b] \mid f(x) \geq t\}$; approximate s from inside and outside by sequences and use continuity.*

Corollary 8.1. *Every polynomial of odd degree has a real root. Every $y \geq 0$ has an n-th root $\sqrt[n]{y}$ (unique); for odd n, every real number has a unique real n-th root.*

Theorem 8.2 (Maximum theorem)**.** *If K is a compact interval and $f : K \to \mathbb{R}$ is continuous, then $f(K)$ is a compact interval: f attains a minimum and a maximum.*

Sketch. *Boundedness: otherwise pick x_n with $f(x_n) > n$; a convergent subsequence (Bolzano–Weierstrass) contradicts continuity. The sup is attained similarly.*

Lemma 8.1. *A positive continuous function on a compact interval is bounded below by a positive constant.*

Theorem 8.3 (Inverse function theorem (continuous version))**.** *If I is an interval and $f : I \to \mathbb{R}$ is injective and continuous, then f is strictly monotone and $f^{-1} : f(I) \to I$ is continuous.*

Corollary 8.2. *The root function $y \mapsto \sqrt[n]{y}$ is continuous on $[0,\infty)$ (all of $\mathbb{R}$ for odd n).*

9 Limits of Functions

Definition 9.1 (Limit of a function)**.** $\lim_{x\to a} f(x) = b$ iff for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - b| < \varepsilon$ for all $x \in I \setminus \{a\}$ with $|x - a| < \delta$. One-sided limits $\lim_{x\to a^-}$, $\lim_{x\to a^+}$; limits at $\pm\infty$: replace $|x - a| < \delta$ by $x > D$ (or $x < D$). Improper limits $\pm\infty$ allowed as values.

Proposition 9.1 (Characterizations of continuity)**.** *For $f : I \to \mathbb{R}$, $a \in I$: f continuous at a iff $\lim_{x\to a} f(x) = f(a)$ iff the one-sided limits both equal $f(a)$.*

Proposition 9.2 (Limit rules)**.** *If $\lim_{x\to a} f(x) = \infty$ then $\lim_{x\to a} 1/f(x) = 0$; if $\lim_{x\to a} f(x) = 0$ and $f > 0$ near a, then $\lim_{x\to a} 1/f(x) = +\infty$. No rule of the form $0 \cdot \infty =?$: e.g. $\lim_{x\to 0^+} cx \cdot \frac{1}{x} = c$ for any c.*

Definition 9.2 (Landau symbols)**.** As $x \to a$: $f = o(h)$ iff $f/h \to 0$; $f = O(h)$ iff f/h stays bounded; $f \sim h$ iff $f/h \to 1$. Useful: $f = g + o(h)$ means equality up to an error that is $o(h)$.

Proposition 9.3 (Landau rules)**.** *$f = O(x^m)$, $g = O(x^n)$ $(x \to 0) \Rightarrow f + g = O(x^{\min(m,n)})$; for $x \to \infty$: $O(x^{\max(m,n)})$. $f = O(h_1)$, $g = O(h_2) \Rightarrow fg = O(h_1 h_2)$; o in one factor gives o in the product. $f \sim g \Rightarrow f = O(g)$ and $g = O(f)$; $\sim$ is an equivalence relation.*

Lemma 9.1. *If $f : [0,\infty) \to \mathbb{R}$ is continuous, $f \geq 0$, and $\lim_{x\to\infty} f(x) = 0$, then f attains a maximum.*

Sketch. *Pick s with $f(s) > 0$; for large x, $f(x) < f(s)$; apply the maximum theorem on $[0, D]$.*

10 Complex Numbers

Definition 10.1 (Complex numbers)**.** $\mathbb{C} = \mathbb{R}^2$ as a set; $z = (x,y) = x + \mathrm{i}y$ with $\operatorname{Re} z = x$, $\operatorname{Im} z = y$, $\mathrm{i}^2 = -1$.

$$z + z' = (x + x', y + y'), \qquad zz' = (xx' - yy',\ xy' + yx').$$

$\mathbb{C}$ is a field. $\overline{z} = x - \mathrm{i}y$ (conjugate), $|z| = \sqrt{x^2 + y^2}$, $z\overline{z} = |z|^2$, $z^{-1} = \overline{z}/|z|^2$ for $z \neq 0$.

Definition 10.2 (Open / connected / domain in $\mathbb{C}$)**.** $X \subseteq \mathbb{C}$ is *open* iff every point has an open disk $U_r(a) = \{z \mid |z - a| < r\}$ inside X. X is *connected* iff any two points can be joined by a continuous path ("polygonal path") in X. *Domain* = open + connected. $X \setminus \{c_1, \ldots, c_s\}$ is a domain for any finite set of points.

Proposition 10.1 (Complex limits)**.** *$\lim z_n = c$ iff $\lim \operatorname{Re} z_n = \operatorname{Re} c$ and $\lim \operatorname{Im} z_n = \operatorname{Im} c$. Cauchy criterion, Bolzano–Weierstrass, and absolute convergence carry over to $\mathbb{C}$; monotonicity, sup/inf, and $\pm\infty$ limits do not.*

Theorem 10.1 (Complex rational functions)**.** *Every rational function $z \mapsto f(z)/g(z)$ ($f, g \neq 0$ complex polynomials) is defined and continuous on the domain $G = \{z \mid g(z) \neq 0\}$. The zeros of g (after cancellation) are the* poles*; at a pole c, $\lim_{z\to c} |h(z)| = \infty$.*

Theorem 10.2 (Partial fraction decomposition)**.** *If $\deg f < \deg g$ and $c_1, \ldots, c_r$ are the zeros of g with multiplicities $e_1, \ldots, e_r$, then*

$$\frac{f(z)}{g(z)} = \sum_{j=1}^{r} \sum_{k=1}^{e_j} \frac{\alpha_{jk}}{(z - c_j)^k}$$

for unique constants $\alpha_{jk} \in \mathbb{C}$.

Example 10.1. $\frac{z^2+z+1}{(z-1)^2(z+2)} = \frac{1}{(z-1)^2} + \frac{2/3}{z-1} + \frac{1/3}{z+2}$. Method: multiply by $(z - c_j)^{e_j}$ and evaluate at c_j to get α_{j,e_j}; subtract and repeat.

11 Power Series

Definition 11.1 (Pointwise / uniform convergence)**.** (f_n) *converges pointwise* to f iff $f_n(x) \to f(x)$ for each x. *Uniformly* iff for every $\varepsilon > 0$ there is D with $|f_n(x) - f(x)| < \varepsilon$ for all x and all $n > D$ (one D works for all x).

Example 11.1. $f_n(x) = x^n$ on $[0,1]$ converges pointwise to 0 for $x < 1$ and 1 at $x = 1$ ğ2014 the limit is discontinuous, so the convergence is not uniform (on $[0,1]$ or $[0,1)$), but it is uniform on every $[0,b]$, $b < 1$.

Theorem 11.1. *A uniform limit of continuous functions is continuous.*

Sketch. *3ε-argument: $|f(x) - f(a)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(a)| + |f_n(a) - f(a)|$.*

Theorem 11.2 (Uniform Cauchy criterion)**.** *A series of functions $\sum f_n$ converges uniformly iff for every $\varepsilon > 0$ there is D with $|\sum_{n=m+1}^{m+k} f_n(x)| < \varepsilon$ for all x, all $m > D$, all k. Uniform absolute convergence ($\sum \|f_n\|_\infty$ converges) implies uniform convergence; a geometric majorant $|f_n(x)| \leq cq^n$ $(q < 1)$ suffices.*

Definition 11.2 (Power series)**.** $\sum_{n=0}^{\infty} a_n (z - a)^n$ is a *power series* about $a \in \mathbb{C}$.

Theorem 11.3 (Convergence of power series)**.** *For a power series about a there is a* radius of convergence *$r \in [0,\infty]$ with*

$$U_r(a) \subseteq K \subseteq D_r(a),$$

where K is the set of convergence points; the series converges absolutely uniformly *on every closed disk $D_\rho(a)$, $\rho < r$. On the boundary circle, anything can happen.*

Corollary 11.1. *The sum of a power series is continuous on its disk of convergence.*

Proposition 11.1 (Operations on power series)**.** *Sums and Cauchy products of power series about the same center are power series about that center (radius at least the minimum of the two). If $\lim |a_n/a_{n+1}|$ exists, it equals the radius of convergence.*

Example 11.2. Radii: $\sum z^n$: $r = 1$ (diverges everywhere on the boundary); $\sum z^n/n$: $r = 1$ (converges on the whole boundary except $z = 1$); $\sum z^n/n^2$: $r = 1$ (converges absolutely on the whole boundary); $\sum n! z^n$: $r = 0$; exp, cos, sin series: $r = \infty$.

12 The Exponential Function

Definition 12.1 (Exponential function)**.** $\exp : \mathbb{C} \to \mathbb{C}$, $\exp z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$ (radius ∞). Write e^z for $\exp z$; $\mathrm{e} = \exp 1 \approx 2.72$.

Theorem 12.1 (Functional equation)**.** $\exp(w + z) = \exp(w)\exp(z)$ *for all* $w, z \in \mathbb{C}$*; consequently* $\exp z \neq 0$ *and* $1/\exp z = \exp(-z)$.

Sketch. *Cauchy product of the two absolutely convergent series + binomial theorem.*

Theorem 12.2 (Real exponential and logarithm)**.** $\exp : \mathbb{R} \to (0, \infty)$ *is strictly increasing and bijective; its continuous inverse is the* logarithm $\log : (0, \infty) \to \mathbb{R}$.

Definition 12.2 (General powers)**.** $a^z := \mathrm{e}^{z \log a}$ for $a > 0$, $z \in \mathbb{C}$.

Proposition 12.1 (Growth comparisons)**.** *For every* $b \in \mathbb{R}$*:* $\lim_{x\to\infty} \mathrm{e}^x/x^b = \infty$*; for* $b > 0$*:* $\lim_{x\to\infty} \log x/x^b = 0$ *and* $\lim_{x\to 0^+} x^b \log x = 0$*. In Landau terms:* $\log x = o(x^b)$ *(*$x \to \infty$*) and* $\log x = o(x^{-b})$ *(*$x \to 0^+$*).*

Theorem 12.3 (Euler's formula)**.** $\mathrm{e}^{\mathrm{i}z} = \cos z + \mathrm{i}\sin z$ *for all* $z \in \mathbb{C}$*, where*

$$\cos z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k}}{(2k)!}, \qquad \sin z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{(2k+1)!}.$$

Hence $\cos z = \frac{\mathrm{e}^{\mathrm{i}z} + \mathrm{e}^{-\mathrm{i}z}}{2}$, $\sin z = \frac{\mathrm{e}^{\mathrm{i}z} - \mathrm{e}^{-\mathrm{i}z}}{2\mathrm{i}}$, *and* $\cos^2 y + \sin^2 y = 1$, $|\mathrm{e}^{\mathrm{i}y}| = 1$ *for real* y.

Theorem 12.4 (Addition theorems)**.** $\cos(x + y) = \cos x \cos y - \sin x \sin y$; $\sin(x+y) = \sin x \cos y + \cos x \sin y$.

Sketch. *Compare real and imaginary parts in* $\mathrm{e}^{\mathrm{i}(x+y)} = \mathrm{e}^{\mathrm{i}x}\mathrm{e}^{\mathrm{i}y}$.

Theorem 12.5 (Periodicity)**.** *The map* $y \mapsto \mathrm{e}^{\mathrm{i}y}$ *wraps* $\mathbb{R}$ *onto the unit circle* $S = \{w \in \mathbb{C} \mid |w| = 1\}$ *counterclockwise;* $\mathrm{e}^{\mathrm{i}x} = \mathrm{e}^{\mathrm{i}y}$ *iff* $x - y \in 2\pi\mathbb{Z}$*. The number* π *is defined by:* $\pi/2$ *is the unique zero of* cos *in* $(0, 2)$*; then* $\mathrm{e}^{\mathrm{i}\pi/2} = \mathrm{i}$, $\mathrm{e}^{\mathrm{i}\pi} = -1$.

Definition 12.3 (Trigonometric / inverse functions)**.** $\tan = \sin/\cos$ on $\mathbb{C} \setminus \{(k + \frac{1}{2})\pi\}$, $\cot = \cos/\sin$ on $\mathbb{C} \setminus \pi\mathbb{Z}$. Restricted inverses: $\arccos : [-1, 1] \to [0, \pi]$, $\arcsin : [-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$, $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$, $\operatorname{arccot} : \mathbb{R} \to (0, \pi)$.

Theorem 12.6. *On any open interval of length* $< 2\pi$*,* $y \mapsto \mathrm{e}^{\mathrm{i}y}$ *is injective with continuous inverse.* $f : \mathbb{R} \to \mathbb{C}$ *is* T*-periodic (*$T > 0$*) iff* $f(t) = g(\mathrm{e}^{2\pi\mathrm{i}t/T})$ *for a unique* $g : S \to \mathbb{C}$*;* f *continuous iff* g *continuous.*

Remark 12.1. $\mathrm{e}^z = \mathrm{e}^x(\cos y + \mathrm{i}\sin y)$ for $z = x + \mathrm{i}y$: $|\mathrm{e}^z| = \mathrm{e}^x$ depends only on x, the "angle" $\mathrm{e}^{\mathrm{i}y}$ only on y – polar coordinates. A single-valued complex logarithm requires restricting exp (e.g. to a strip $\{z \mid -\pi < \operatorname{Im} z \leq \pi\}$, giving the principal branch Log with $\operatorname{Log} w = \log|w| + \mathrm{i}\operatorname{Arg} w$, $\operatorname{Arg} \in (-\pi, \pi]$); then $\sqrt[n]{z} := \mathrm{e}^{\operatorname{Log} z/n}$ is one of n possible root functions.

13 Differentiation

Definition 13.1 (Derivative)**.** $f : I \to \mathbb{R}$ is *differentiable at* $a \in I$ iff

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

exists in $\mathbb{R}$. Equivalent "little-o" form: $f(a + h) = f(a) + bh + o(h)$ as $h \to 0$ iff $f'(a) = b$.

Proposition 13.1. *Differentiable* $\Rightarrow$ *continuous. Not conversely:* $f(x) = |x|$ *is continuous but not differentiable at* 0*. There even exist continuous functions differentiable nowhere.*

Proposition 13.2 (Differentiation rules)**.** $(f + g)' = f' + g'$; $(\lambda f)' = \lambda f'$; $(fg)' = f'g + fg'$; $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$; *chain rule:* $(g \circ f)' = (g' \circ f) \cdot f'$.

Proposition 13.3 (Standard derivatives)**.** $(x^n)' = nx^{n-1}$ *(all* $n \in \mathbb{Z}$*, on the natural domain);* $(\mathrm{e}^x)' = \mathrm{e}^x$; $(\log x)' = 1/x$ *(*$x > 0$*);* $(\sqrt[n]{x})' = \frac{1}{n}x^{\frac{1}{n}-1}$ *(*$x > 0$*);* $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$; $(\tan x)' = 1 + \tan^2 x = 1/\cos^2 x$; $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$; $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$; $(\arctan x)' = \frac{1}{1+x^2}$; $(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$; $(x^b)' = bx^{b-1}$ *(*$x > 0$*,* $b \in \mathbb{R}$*).*

Definition 13.2 (Higher derivatives / C^k)**.** $f^{(k)} = (f^{(k-1)})'$; f is *k-times differentiable*; if $f^{(k)}$ is continuous, f is a *C^k-function.* C^0 = continuous, C^1 = differentiable with continuous derivative, C^∞ = smooth.

Remark 13.1. Complex differentiability of $f : G \to \mathbb{C}$ ($G \subseteq \mathbb{C}$ a domain) means the complex limit $f'(a) = \lim_{z\to a} \frac{f(z)-f(a)}{z-a}$ exists – a much stronger condition than real differentiability (the map $z \mapsto \overline{z}$ is nowhere complex-differentiable). On domains, complex-differentiable functions are automatically C^∞.

14 The Mean Value Theorem

Theorem 14.1 (Rolle's theorem)**.** *If* $f : [a, b] \to \mathbb{R}$ *is continuous, differentiable on* (a, b)*, and* $f(a) = f(b)$*, then there is* $t \in (a, b)$ *with* $f'(t) = 0$.

Sketch. *At an interior maximum (or minimum), the difference quotients have opposite signs from left and right.*

Theorem 14.2 (Mean value theorem)**.** *If* $f : [a, b] \to \mathbb{R}$ *is continuous and differentiable on* (a, b)*, then there is* $t \in (a, b)$ *with*

$$\frac{f(b) - f(a)}{b - a} = f'(t).$$

Sketch. *Apply Rolle to* $h(x) = f(x) - \alpha - \beta x$ *with* $\beta = \frac{f(b)-f(a)}{b-a}$ *chosen so* $h(a) = h(b)$.

Corollary 14.1. *If* $f' = 0$ *on an interval, then* f *is constant.*

Theorem 14.3 (Monotonicity)**.** *For* $f : [a, b] \to \mathbb{R}$ *continuous, differentiable on* (a, b)*:*

- $f'(x) \geq 0$ *on* $(a, b) \Leftrightarrow f$ *increasing on* $[a, b]$*;*
- $f'(x) > 0$ *on* $(a, b) \Rightarrow f$ *strictly increasing on* $[a, b]$ *(converse false:* x^3*).*

Theorem 14.4 (Derivative of the inverse)**.** *If* $f : I \to \mathbb{R}$ *is differentiable with* $f' > 0$ *everywhere, then* f *is strictly increasing and* $f^{-1} : f(I) \to I$ *is differentiable with*

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Definition 14.1 (Local extremum)**.** f has a *local minimum* at a iff $f(x) \geq f(a)$ for all x near a; *strict* if $>$ for $x \neq a$. Dually for maximum.

Theorem 14.5 (Extremum criteria)**.** *Let* I *be open,* $f : I \to \mathbb{R}$ *differentiable.*

- *If* f *has a local extremum at* a*, then* $f'(a) = 0$ *(necessary condition).*

- *If $f'(a)=0$ and $f''(a)>0$, then f has a strict local minimum at a (dually $f''(a)<0$: strict local maximum). If $f''(a)=0$: no conclusion (x^4 vs. x^3 at 0).*

Theorem 14.6 (L'Hôpital's rule $(\frac{0}{0})$)**.** *If $f,g:(a,b)\to\mathbb{R}$ are differentiable, $\lim_{x\to b} f(x)=\lim_{x\to b} g(x)=0$, $g'\neq 0$, and $\lim_{x\to b} f'(x)/g'(x)$ exists (possibly $\pm\infty$), then $\lim_{x\to b} f(x)/g(x)$ exists and equals it. Analogous $\frac{\infty}{\infty}$ version. Also valid for one-sided and two-sided limits.*

Example 14.1. $\lim_{n\to\infty}\left(1+\frac{t}{n}\right)^n = \mathrm{e}^t$: write as $\exp\big(x\log(1+t/x)\big)$ and apply L'Hôpital to $\frac{\log(1+t/x)}{1/x}$.

15 Analytic Functions

Definition 15.1 (Analytic)**.** $f:X\to\mathbb{C}$ ($X\subseteq\mathbb{R}$ interval or $X\subseteq\mathbb{C}$ domain) is *analytic* iff every $a\in X$ has a neighborhood on which f equals a convergent power series $\sum a_n(z-a)^n$.

Theorem 15.1 (Composition of power series)**.** *If $f(z)=\sum a_j(z-a)^j$ converges on $U_\delta(a)$ and $f\big(U_\delta(a)\big)\subseteq U_r(b)$ where $g(w)=\sum b_k(w-b)^k$ has radius r, then $g\circ f$ is given by a convergent power series about a on $U_\delta(a)$.*

Proposition 15.1. *Sums, products, quotients (denominator nonzero at the point), and compositions of analytic functions are analytic. Every power series defines an analytic function on its disk of convergence (re-expand about any interior point).*

Theorem 15.2 (Termwise differentiation)**.** *A power series and its termwise derivative have the same radius of convergence, and on the disk of convergence*

$$\frac{\mathrm{d}}{\mathrm{d}z}\sum_{n=0}^{\infty}a_n(z-a)^n=\sum_{n=1}^{\infty}na_n(z-a)^{n-1}.$$

In particular, every analytic function is C^∞. Termwise integration is also allowed (the integrated series has the same radius).

Example 15.1 (Derivatives from series)**.** $\exp'=\exp$; $(\log x)'=1/x$; $(x^b)'=bx^{b-1}$; $\cos'=-\sin$, $\sin'=\cos$; $(\arctan x)'=\frac{1}{1+x^2}$, $(\arcsin x)'=\frac{1}{\sqrt{1-x^2}}$, $(\arccos x)'=-\frac{1}{\sqrt{1-x^2}}$.

Definition 15.2 (Antiderivative)**.** F is an *antiderivative* (primitive) of f on an interval iff $F'=f$. Any two antiderivatives differ by a constant. For a power series, the termwise integrated series is an antiderivative.

Corollary 15.1. *If f' is analytic on an interval, then f is analytic. Hence* log, arctan, arccot, x^b *($x>0$),* arcsin, arccos *are analytic on their domains. Not analytic: $|x|$ (at 0), $\sqrt[n]{x}$ (at 0, $n>1$),* arcsin / arccos *(at ±1), $z\mapsto\overline{z}$, and the smooth but non-analytic $f(x)=0$ for $x\le 0$, $f(x)=\mathrm{e}^{-1/x}$ for $x>0$ (all derivatives vanish at 0, yet $f\neq 0$).*

16 Taylor Series

Lemma 16.1 (Taylor coefficients)**.** *If an analytic f is represented near a by $\sum a_n(z-a)^n$, then necessarily*

$$a_n=\frac{f^{(n)}(a)}{n!}.$$

The series $\sum\frac{f^{(n)}(a)}{n!}(z-a)^n$ is the Taylor series *of f about a.*

Theorem 16.1 (Identity theorem)**.** *If two analytic functions on an interval/domain have the same Taylor series at one point, they are equal everywhere.*

Example 16.1 (Important Taylor series (about 0))**.**

$$\mathrm{e}^z=\sum_{n=0}^{\infty}\frac{z^n}{n!}\quad(r=\infty),\qquad\frac{1}{1-z}=\sum_{n=0}^{\infty}z^n\quad(r=1),$$

$$\cos z=\sum_{k=0}^{\infty}\frac{(-1)^kz^{2k}}{(2k)!},\quad\sin z=\sum_{k=0}^{\infty}\frac{(-1)^kz^{2k+1}}{(2k+1)!}\quad(r=\infty),$$

$$\log(1+z)=\sum_{m=1}^{\infty}\frac{(-1)^{m-1}}{m}z^m\quad(r=1),$$

$$\arctan z=\sum_{m=0}^{\infty}\frac{(-1)^m}{2m+1}z^{2m+1}\quad(r=1),$$

$$(1+z)^b=\sum_{n=0}^{\infty}\binom{b}{n}z^n\quad(r=1).$$

Here $\binom{b}{n}=\frac{b(b-1)\cdots(b-n+1)}{n!}$ is the *generalized binomial coefficient*. The last series is the *binomial series*; for $b\in\mathbb{N}$ it terminates (binomial formula, $r=\infty$). In particular $\sqrt{1+z}=1+\frac{z}{2}-\frac{z^2}{8}+\frac{z^3}{16}-\frac{5z^4}{128}+\cdots$.

Example 16.2 (Taylor series of rational functions)**.** Expand $\frac{1}{z-c}$ about $a\neq c$ as a geometric series in $\frac{z-a}{c-a}$: valid for $|z-a|<|c-a|$. Differentiating gives the series for $\frac{1}{(z-c)^n}$. The Taylor series of a rational function about a has radius exactly $\min_j|a-c_j|$ over the poles c_j. Example: $\frac{1}{1+x^2}=\sum(-1)^nx^{2n}$ has radius 1 ☹2014 the complex poles $\pm$i explain this from the real viewpoint.

Theorem 16.2. *If f is analytic on $U_r(a)$, its Taylor series about a has radius at least r.*

Definition 16.1 (Taylor polynomial)**.** For a C^k-function f on an interval,

$$T_a^kf(x)=\sum_{j=0}^{k}\frac{f^{(j)}(a)}{j!}(x-a)^j$$

is the *k-th Taylor polynomial* at a.

Proposition 16.1 (Best approximation property)**.** *Among all polynomials of degree $\le k$, T_a^kf is the unique one with*

$$f(x)=T_a^kf(x)+o(|x-a|^k)\quad(x\to a),$$

i.e. it matches f and its first k derivatives at a: $\big(T_a^kf\big)^{(j)}(a)=f^{(j)}(a)$ for $j\le k$.

Proposition 16.2 (Computing with Taylor polynomials)**.** *$T_a^k(f+g)=T_a^kf+T_a^kg$; $T_a^k(f\cdot g)=T_a^k\big(T_a^kf\cdot T_a^kg\big)$ (truncate the product to degree k); $T_a^k(g\circ f)=T_a^k\big(T_{f(a)}^kg\circ T_a^kf\big)$ (truncate). For $k=1$ the composition rule is the chain rule.*

Remark 16.1 (Caution). The Taylor series of a C^∞-function need not converge to the function: for $f(x)=0$ ($x\le0$), $\mathrm{e}^{-1/x}$ ($x>0$), the Taylor series at 0 is the zero series, yet $f\neq0$.

Definition 16.2 (Asymptotic expansion)**.** $f=\sum_{k=0}^{n}\lambda_kg_k+o(g_n)$ as $x\to a$, where $g_{k+1}=o(g_k)$, is an *asymptotic expansion* of f w.r.t. the scale (g_k). Taylor series are the special case $g_k(x)=(x-a)^k$. The series need not converge.

17 Quick Reference: Limits

- $\lim_{n\to\infty} \sqrt[n]{n} = 1$, $\lim_{n\to\infty} \sqrt[n]{n!} = \infty$, $\lim_{n\to\infty} \frac{n!}{n^n} = 0$.
- $\lim_{n\to\infty} \frac{a^n}{n!} = 0$ for every fixed a; $\lim_{n\to\infty} \frac{n!}{a^n} = \infty$.
- $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = \mathrm{e}^x$; $\left(1 + \frac{1}{n}\right)^n \uparrow \mathrm{e}$, $\mathrm{e} < 3$.
- $\lim_{x\to 0} \frac{\sin x}{x} = 1$; $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$; $\lim_{x\to 0} \frac{\mathrm{e}^x - 1}{x} = 1$; $\lim_{x\to 0} \frac{\log(1+x)}{x} = 1$.
- $\lim_{x\to\infty} \frac{\log x}{x^b} = 0$ $(b > 0)$; $\lim_{x\to\infty} \frac{\mathrm{e}^x}{x^b} = \infty$ (any b).
- $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} f(k/n) = \int_0^1 f(x)\,\mathrm{d}x$ (Riemann sums).
- $\sum_{n=1}^{\infty} \frac{1}{n^d}$ converges iff $d > 1$; $\sum \frac{(-1)^{n-1}}{n} = \log 2$.
- $\sum_{n=0}^{\infty} \frac{1}{n!} = \mathrm{e}$; $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} = \sin 1$; $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} = \cos 1$.

18 Quick Reference: Inequalities

- Triangle: $|x \pm y| \le |x| + |y|$; reverse: $|x+y| \ge \big||x| - |y|\big|$.
- Bernoulli: $(1+x)^n \ge 1 + nx$ for $x \ge -1$, $n \in \mathbb{N}$.
- AM–GM: $\frac{x_1 + \cdots + x_n}{n} \ge \sqrt[n]{x_1 \cdots x_n}$ for $x_i \ge 0$.
- Cauchy–Schwarz: $\left|\sum x_i y_i\right| \le \sqrt{\sum x_i^2}\sqrt{\sum y_i^2}$.
- $|\sin x| \le |x|$; $|1 - \cos x| \le \frac{x^2}{2}$; $\mathrm{e}^x \ge 1 + x$ for all x.
- $\log x \le x - 1$ for $x > 0$, with equality iff $x = 1$; $\log(1+x) \le x$.